

Neuroscientific Frontiers — Module 04: Cognition —

Study Questions

1. How does the prefrontal cortex implement hierarchical policy evaluation? Map dlPFC, vmPFC, ACC, and frontopolar cortex to specific Active Inference computations.
2. How is expected free energy (EFE) neurally decomposed into pragmatic and epistemic components? What brain regions encode each component?
3. What is the neural signature of decision difficulty? How does ACC activation reflect uncertainty about policy selection?
4. How does frontopolar cortex (BA10) implement counterfactual reasoning? Why is maintaining an "alternative strategy in the background" computationally valuable?
5. Explain how tonic vs. phasic dopamine map onto policy precision vs. reward prediction errors. What happens when tonic DA is too high (schizophrenia) vs. too low (Parkinson's)?
6. What is the gamma parameter in Active Inference? How does it map onto dopaminergic function? What behavioral signatures indicate high vs. low gamma?
7. How does working memory function as active maintenance of beliefs about unobserved hidden states? What neural activity corresponds to belief maintenance?
8. Explain theta-gamma coupling in working memory. How does the number of gamma bursts per theta cycle explain working memory capacity limits (~4 items)?
9. What is hippocampal replay? How does it implement mental simulation of candidate policies during planning? What evidence shows that replay corresponds to prospective evaluation?
10. What is prospective coding in hippocampal place cells? How does "looking ahead along planned routes" implement T-step policy evaluation?
11. How do prefrontal-hippocampal theta interactions support planning? What would happen to planning ability if this coupling were disrupted?
12. Compare the Active Inference account of exploration-exploitation with the UCB1 algorithm and Thompson sampling from reinforcement learning. What are the computational differences?

13. How does the ACC "conflict monitoring" model (Botvinick et al.) relate to Active Inference's treatment of policy uncertainty? Are these equivalent frameworks or genuinely different?
14. What is the neural evidence for separate "curiosity" (epistemic) and "reward" (pragmatic) signals? How do anterior insula and vmPFC contribute differently?
15. How does Active Inference explain sunk cost bias? When should an agent abandon a policy, and what neural mechanism signals policy switching?
16. How does prefrontal damage impair planning? Map different types of prefrontal lesions to specific Active Inference deficits (policy generation, evaluation, sequencing).
17. How do orbitofrontal cortex value representations relate to the C vector (preferences) in Active Inference? What happens to preferences after OFC damage?
18. What role does the anterior cingulate play in foraging decisions (patch leaving)? How does expected free energy determine when to leave a depleted resource?
19. How might chronic stress alter policy evaluation? What happens to exploration when the allostatic system is overloaded (connecting to Module 08)?
20. Critically evaluate: Does Active Inference provide a better account of decision-making than prospect theory or dual-process theory? What predictions uniquely distinguish Active Inference from these alternatives?